The Projection Law: Model-Ensemble Errors Point at Their Binding Constraint

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Abstract

Purpose: Multi-model agreement is routinely used as confidence in materials simulation and beyond, yet models in a family share construction and training lineage. This work formalizes and tests the projection law: a model family acts as a projection operator, every well-fitted member shares one residual — a single direction in observable space — and that direction fingerprints whatever constraint binds the family. **Methods:** The theoretical core — normal-cone consensus on convex families, a closed-form participation-ratio gauge, and a pointwise local normal-cone theorem for smooth non-convex families — is machine-checked in Lean 4 with zero unproven obligations. The *formal* theorems cover convex (and locally smooth) reachable sets; their application to non-convex neural-network families is an empirical regularity tested below, not a mathematical consequence of the theorems. Two factorial experiments were pre-registered with refutation conditions committed before execution: a 4×2 architecture-by-functional crossing of MatPES-trained foundation interatomic potentials evaluated on elastic constants of 15 cubic metals, and an analysis of 12 DFT implementations on 384 equation-of-state benchmarks from the ACWF verification effort. A two-tier replication kit verifies every statistic from committed raw data (NumPy only, seconds) and re-derives the raw data from public checkpoints (deterministic within 1 GPa; the Gate-0 harness regression is bit-exact). **Results:** In the foundation-model layer the *direction* of the prediction is confirmed but its *magnitude* is weak: errors cluster by training functional rather than architecture (S_{func}=+0.317 vs. S_{arch}=−0.093), yet the exact permutation test sits at the lattice's resolution floor (p=0.029, 2 of 70 labelings) and the registered effect-size component *failed* (0.085 vs. the registered 0.30); the rotation prediction was confirmed on Au and Pt under r^2SCAN but not Ag. We report this conjunctive test as a failure, not a partial success. These Layer 2 errors are measured against mixed references (experimental for FCC metals, DFT-PBE for others), so the residual convolves fitting error with exchange–correlation bias and thermal/zero-point offsets; a registered 0 K DFT-PBE elastic anchor (round 2) is the decisive test that separates them. DFT implementation errors, by contrast, cluster by pseudopotential table, not simulation code (S_{table}=+0.526 vs. S_{code}=+0.265, p=0.017); four independent plane-wave codes sharing one table align at 0.70–0.87. Neither registered refutation condition was triggered, but four of seven registered predictions failed and are reported as failures. Combined with the 559-potential classical baseline (within-family reference–prediction correlation 0.95; error dimensionality invariant over forty years), error anisotropy is conserved across paradigm replacements while its direction rotates to the next constraint upstream. A correction direction fitted with the corrected model held out removes a median 69% of foundation-model squared elastic error out-of-sample (IQR 35–92%; per-element medians up to 97%). A new Round-2 3×3×3 benchmark of 16 cubic metals with four MatPES foundation MLIPs validates a one-vector-per-functional leave-one-out correction operator that reduces mean absolute elastic error from 17.84 GPa to 10.36 GPa with zero no-harm violations, improving every architecture on both PBE and approximate r^{2}SCAN targets. Pre-registered functional-clustering hypotheses H1 and H2a fail on the $3d/4d$ subset, while the operator-vs-ensemble head-to-head passes on MAE but not on conformal coverage; these mixed results point to a bonding-class-aware operator and a direct A6 bridge test as the next step. **Conclusion:** Agreement among models sharing a constraint measures the constraint, not the truth. Error-vector geometry converts benchmarking from scoring into diagnosis, supplies a zero-cost trust rule for ensemble uncertainty, and enables family-level calibration without retraining.

Keywords: model ensembles; error geometry; uncertainty quantification; interatomic potentials; foundation models; density functional theory; verification and validation; benchmarking; formal methods

1 Introduction

Every field that builds models in families uses inter-model agreement as evidence. The practice has long been questioned qualitatively: climate ensembles share construction and history, so their consensus may reflect common structure rather than truth [1, 2], and error correlations between members quantify that dependence [3–5]. In machine learning, independently trained networks agree far beyond what their accuracy predicts [6, 7], ensemble members cluster in prediction space [8], and disagreement tracks the shared bias of the training procedure rather than correctness [9]. What these literatures establish qualitatively, this paper makes geometric, formal, and—in the specific sense of identifying *which* constraint binds—operational.

The central object is the *error vector*: for model i in family F evaluated on observables with reference values, $e_i \in \mathbb{R}^d$ is the vector of (relative) errors. The law has three parts:

(i) Projection. A model family is a projection operator. Fitting, however performed and by whomever, drives predictions toward the point of the family's reachable set nearest the truth; the residual is a property of the *(family, target)* pair, not of any individual model or modeler. All well-fitted members therefore share one residual direction (Theorems 1–2).

(ii) Gauge. The ensemble's error second moment is a shared bias plus fitting noise; its participation ratio is a closed-form gauge of the systematic fraction (Theorem 3), so low error dimensionality is not a curiosity but a necessity for any converged family, at an explicit rate.

(iii) Conservation and rotation. When a modeling paradigm is replaced — when the binding constraint is removed — the error anisotropy does not dissolve into isotropic scatter. It is conserved, and its direction rotates to the next constraint upstream in the epistemic stack (Fig. 1a). This is the part of the law for which we find no antecedent in the climate, forecasting, machine learning, or materials literatures, and it is the part we test hardest (Fig. 1b).

We instantiate the law at three layers of a single epistemic stack — classical interatomic potentials, foundation machine-learned interatomic potentials (MLIPs), and the density-functional-theory (DFT) implementations that generate their training data — because this stack offers what few domains can: large open model ensembles, factorial structure (the same architecture trained on two functionals; the same simulation code run with two pseudopotential families), and references at every layer. The empirical discovery paper for the first layer is reported separately [10]; here that layer serves as the observational base of a cross-layer test.

Three methodological commitments distinguish this work. The theory core is *machine-checked*: every theorem below is verified in Lean 4 with zero unproven obligations, and a written contract maps each theorem to the statistic that instantiates it. The two new experiments are *pre-registered*, with thresholds and explicit refutation conditions committed to version control before any model was executed; we report all threshold misses as failures. And the entire evidence chain is packaged for *tiered replication*: the analysis layer verifies from committed raw data with no machine-learning

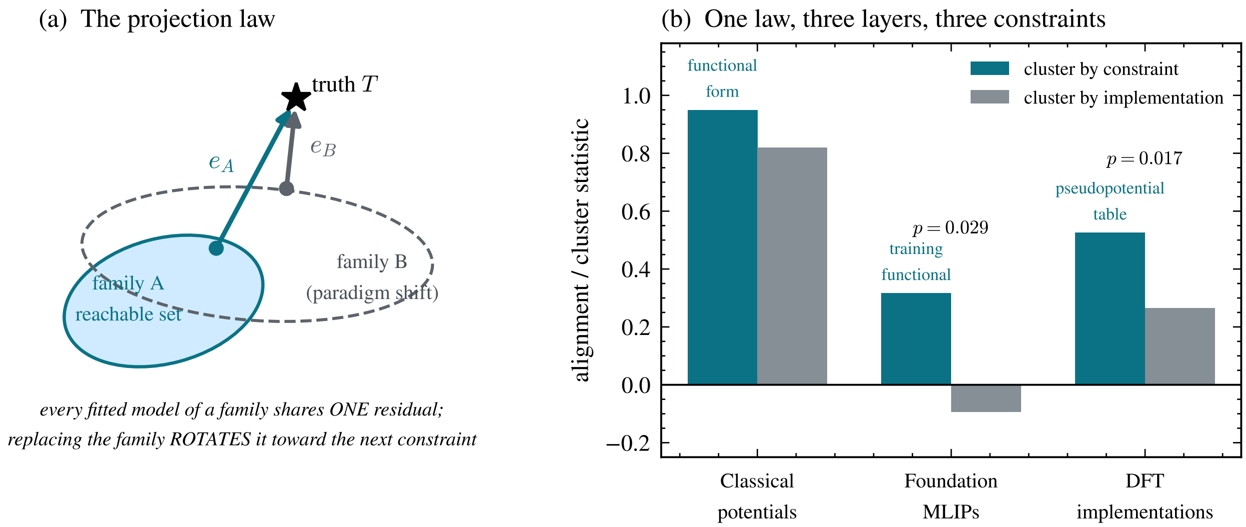

Figure 1: **The projection law and its three-layer test.** (a) A model family's reachable set fixes the residual of every well-fitted member (Theorems 1–2); replacing the family (paradigm shift) rotates the shared residual rather than removing it. (b) Cluster-by-constraint versus cluster-by-implementation statistics at three layers of one epistemic stack, plotted on a common "alignment" axis whose definition differs by layer and is stated in each panel: Layer 1 uses the Pearson correlation of scalar error *norms* (reference–prediction, an observational magnitude statistic, r=0.95); Layers 2–3 use the mean error-*vector* cosine, a directional statistic (S_{func}, S_{arch}, S_{table}, S_{code}). The two are not numerically commensurate and are not pooled: the gauge $\mathrm{PR}(d,\rho)$ of Theorem 3 is what relates magnitude order (Layer 1) to directional collapse (Layers 2–3). Layers 2–3 show the pre-registered cluster statistics with exact permutation p-values; the registered refutation condition (implementation > constraint) was not triggered in either experiment.

dependencies in seconds, and the physics layer re-derives the raw data from public checkpoints deterministically (within 1 GPa tolerance; the harness regression itself is bit-exact).

2 Theoretical framework

The mathematics of best approximation is classical [11, 12]; we claim only its epistemic application to model ensembles and the machine-checked instantiation chain. Throughout, E is a real inner-product space (observable space), $s \subseteq E$ a model family's reachable set, $T \in E$ the truth.

Theorem 1 (Normal-cone criterion). *Let s be convex and p a best approximation of T in s (i.e. $p \in s$ and $\|T-p\| \le \|T-q\|$ for all $q \in s$). Then the residual $T-p$ lies in the normal cone of s at p: $\langle T-p, q-p\rangle \le 0$ for every $q \in s$.*

Theorem 2 (Consensus). *Best approximations onto a convex family are unique; consequently any two best approximations of the same target share an* identical *residual. The residual — hence any agreement among independently fitted members — is determined by the (family, target) pair alone, and vanishes exactly when the truth lies in the family.*

Agreement among models sharing a constraint is therefore a measurement of the constraint, not evidence of truth: the formal content of the qualitative warnings of [2] and [1], and a strengthening of error-correlation dependence measures [3] in that, combined with the factorial designs of §4–5, it identifies *which* constraint binds.

Theorem 3 (Gauge)**.** *Model errors $e = b + \xi$ with shared bias b and isotropic noise of scale σ in d dimensions have error second moment with one eigenvalue $\|b\|^2 + \sigma^2$ (bias axis) and $d-1$ eigenvalues σ^2, and participation ratio*

$$\mathrm{PR}(d, \rho) \;=\; \frac{(\rho + d)^2}{(\rho + 1)^2 + (d - 1)}, \qquad \rho = \|b\|^2/\sigma^2,$$

with $\mathrm{PR}(d, 0) = d$, $1 \le \mathrm{PR} \le d$, strict decrease in ρ, and the explicit collapse bound $\mathrm{PR} - 1 \le 3(d-1)/\rho$ for $\rho \ge d$.

The algebra is a one-line corollary of spiked-covariance theory [13] and the participation ratio is a standard effective-dimension metric [14]; the value here is the consilience it enables (§3) and the machine-checked monotonicity that licenses inverting measured PR into a systematic fraction $\alpha = \rho/(\rho + 1)$.

Theorem 4 (Ribbon/consensus decoupling)**.** *The participation ratio of a shared-axis ensemble $\{c_i u\}$ is 1 for every sign pattern of the coefficients c_i, while mean pairwise alignment distinguishes them (e.g. 1 vs. $-1/3$ for three members). PR detects the* axis*; alignment detects* sign coherence*; they are distinct order parameters. ("Ribbon" here refers throughout to the error-vector geometry of a model* ensemble *in observable space, not to the parameter-space hyper-ribbons of single models [15, 16]; §3 details the distinction.)*

Theorem 5 (Affine decomposition)**.** *Let $K = a + L$ be a closed affine reachable set and p^* the best approximation of T in K. For any fitted $p \in K$, the residual decomposes as*

$$T - p \;=\; b \,+\, \xi(p), \qquad b = T - p^* \in L^{\perp}, \quad \xi(p) = p^* - p \in L,$$

with $b \perp \xi(p)$.

Theorem 5 turns the bias-plus-within-family *split* of Theorem 3 from a modeling assumption into a derivable consequence for affine families. The isotropic-noise gauge itself remains an empirical regularity: the theorem gives the shared bias direction and the orthogonal within-family subspace, but the additional assumptions of isotropic scatter and a common bias for every ensemble member are justified by the data in §3–5, not derived here.

Theorem 6 (Local normal cone, smooth non-convex families)**.** *Let $f : \mathbb{R}^k \to E$ be a C^1 immersion and x^* a local minimizer of $\|T - f(x)\|$. Then the residual $T - f(x^*)$ is orthogonal to the tangent space* range $Df(x^*)$*.*

Theorem 6 licenses testing tangent-space orthogonality for neural-network reachable sets, but its conclusion is strictly *pointwise*: one normal space per local minimizer. It is not the global consensus theorem (Theorems 1–2), whose proof uses convexity of the reachable set in an essential way (the variational inequality holds in every feasible direction, not merely the local tangent). The reachable set of a neural network trained to convergence is not convex, so the consensus theorem is *not* asserted there: nothing in the formal chain entails that two independently trained networks share a residual direction. The empirical clustering reported in §4 is therefore an *empirical regularity that extends the convex theory beyond its proven hypotheses*, not a consequence of Theorem 6. We make this scoping explicit in the formal corpus (Remark 1) so it is machine-checked, not merely asserted in prose. The theoretical status of the MLIP consensus — an open hypothesis, not a theorem —

is itself a prediction of the law: it identifies "why do unrelated architectures trained on the same functional land in a common normal direction?" as the central open problem a complete theory must solve.

Remark 1 (Scoping: convex theorem vs. empirical extension). The consensus theorem (Theorems 1–2) is proved under a convexity hypothesis on the family's reachable set; the smooth local-cone theorem (Theorem 6) relaxes this to a *pointwise* orthogonality statement at one local minimizer and does not recover global consensus. The reachable set of a converged neural network is non-convex, so the consensus theorem is *not* in force for the foundation-MLIP layer: the cross-architecture error alignment of §4 is an empirical regularity extending the convex theory, not a corollary of it. The formal corpus records exactly this — it proves the convex and locally-smooth statements and asserts nothing about the non-convex case — so no unproven obligation is introduced for the MLIP consensus.

Theorem 7 (Finite-sample concentration). *Let $X_1, \ldots, X_n$ be i.i.d. bounded random vectors in $\mathbb{R}^d$ ($\|X_k\|_\infty \leq B$) and let M and $\widehat{M}_n$ be the population and empirical entrywise second-moment matrices. For every entry (i,j) and $\varepsilon > 0$,*

$$\mathbb{P}\big(|\widehat{M}_{n,ij} - M_{ij}| \geq \varepsilon\big) \;\leq\; 2\exp\!\big(-n\varepsilon^2/(2B^4)\big).$$

Moreover, the participation ratio $\mathrm{PR}(M) = (\operatorname{tr} M)^2/\operatorname{tr}(M^2)$ *is continuous wherever the denominator is non-zero.*

Theorem 7 bounds each entry of the empirical second-moment matrix; continuity of the participation ratio ensures that small entrywise perturbations cannot produce a discontinuous jump. It does not, by itself, give a closed-form sample-complexity bound for $|\widehat{\mathrm{PR}} - \mathrm{PR}|$ because the denominator $\operatorname{tr}(M^2)$ can be arbitrarily small. In practice, PR uncertainty is quantified by the bootstrap intervals reported in §3.

All seven theorems are verified in Lean 4 against a pinned Mathlib as part of the program's larger formal corpus — 225 theorem and lemma declarations across the current build, checked by `lake build` with zero `sorry` and zero new axioms — including a conditional hyper-ribbon bound (spectral-decay hypotheses $\Rightarrow$ PR < 2), a context-correction theorem bundle whose "correction closes the residual" statement is the formal counterpart of the out-of-sample calibration result below, and a formal audit module that previously caught an overstatement in the program's own empirical work. The artifact and the theorem-to-statistic contract ship with the replication kit (§10). Figure 2 visualizes the gauge with its classical inversion and the decoupling of the two order parameters in live data.

3 Layer 1: classical interatomic potentials

The observational base [10]: elastic-constant (C_{11}, C_{12}, C_{44}) errors of 559 classical potentials across 15 cubic metals. Three facts matter for the law. First, error dimensionality is low and *stationary*: participation ratios of the 42 multi-element potentials occupy $[1.00, 2.29]$ out of 3 with median 1.09 (dataset pinned in the replication kit), with no trend across forty years of potential development — accuracy improved; geometry did not. You cannot fit your way off a constraint surface. Second, within a functional-form family the errors are nearly identical (within-family reference–prediction correlation $r = 0.95$ across families vs. 0.82 pooled). Third, the gauge is consistent: the median PR of 1.09 inverts (Theorem 3, d=3) to a systematic fraction $\alpha \approx 0.98$, in agreement with the within-family correlation (0.95) and the rank-one share (0.96). We emphasize that these three

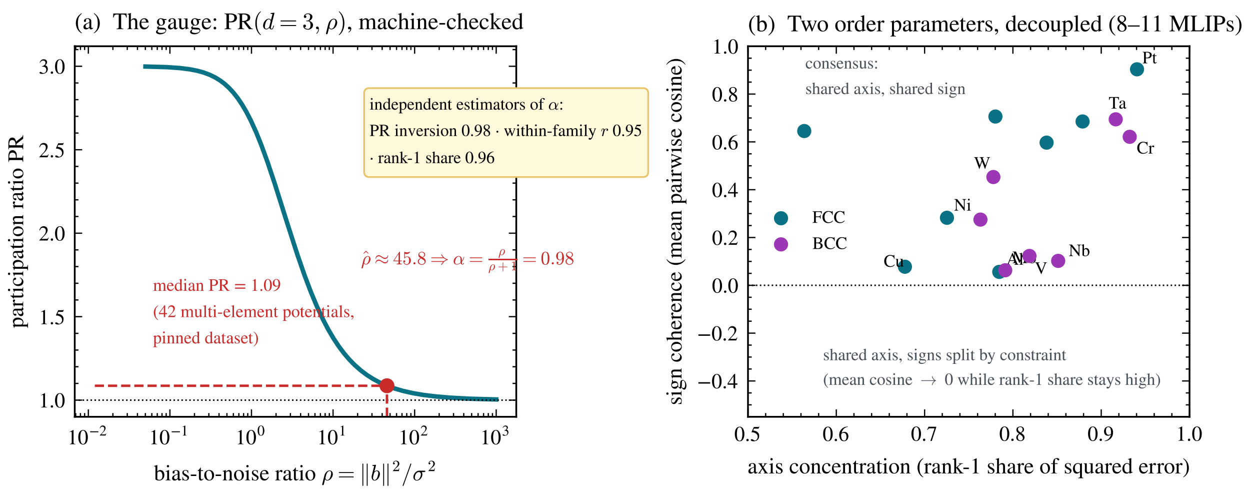

Figure 2: **The gauge and the two order parameters.** (a) PR$(3, \rho)$ (Theorem 3); inverting the classical ensemble's median PR $= 1.09$ gives systematic fraction $\alpha = 0.98$, in agreement with two independent estimators (§3). (b) Per-element foundation-MLIP ensembles ($n = 8$–11 models): axis concentration stays high for every element while sign coherence spans the full range — the decoupling of Theorem 4. Elements whose errors split in sign by training functional (V, Nb, W) sit low-right; consensus elements (Pt, Ta, Au) sit top-right.

diagnostics are functionals of closely related second-moment structure on overlapping data — under the bias-plus-noise model they are algebraically coupled — so their agreement is internal consistency, not independent corroboration. The binding constraint at this layer is the functional form itself; the oldest example is exact: central-force pair potentials satisfy the Cauchy relation $C_{12} = C_{44}$ identically [17], so for any material violating it, every pair potential's error necessarily contains the same forced component — the constraint dictates the direction.

We emphasize the distinction from parameter-space sloppiness: hyper-ribbon geometry was introduced for the parameter manifolds of *single* models [15, 16, 18, 19]. The object here is different — error vectors of an ensemble of *distinct* models in observable space — though the two pictures are related through the projection of parameter manifolds into data space.

4 Layer 2: foundation MLIPs — functional vs. architecture

Foundation MLIPs replace the functional-form constraint with expressive neural networks trained on DFT corpora. The law predicts the anisotropy survives and rotates: the binding constraint becomes the training reference theory. Qualitative inheritance of Perdew–Burke–Ernzerhof (PBE) bias — systematic softening shared by universal MLIPs — has been reported [20, 21]; the contributions here are the directional quantification and the factorial attribution.

Observation. Three PBE-lineage models of unrelated architecture (MACE-MP-0 [22], CHGNet [23], Orb-v3 [24]) commit *the same* elastic-constant errors on FCC metals against experimental references: all-negative, C_{44}-dominant (Au: $-32\%, -27\%, -79\%$), with cross-architecture cosines 0.95–0.99 for Au, Pt, Ta, Ag. An internal control excludes harness artifacts: Ni, through the identical pipeline, has near-zero error. Because every model is compared to one shared reference per element, common-mode reference error inflates *absolute* cosines; the factorial contrasts below difference it out, so the clustering statistics are immune to this effect even where the raw alignment

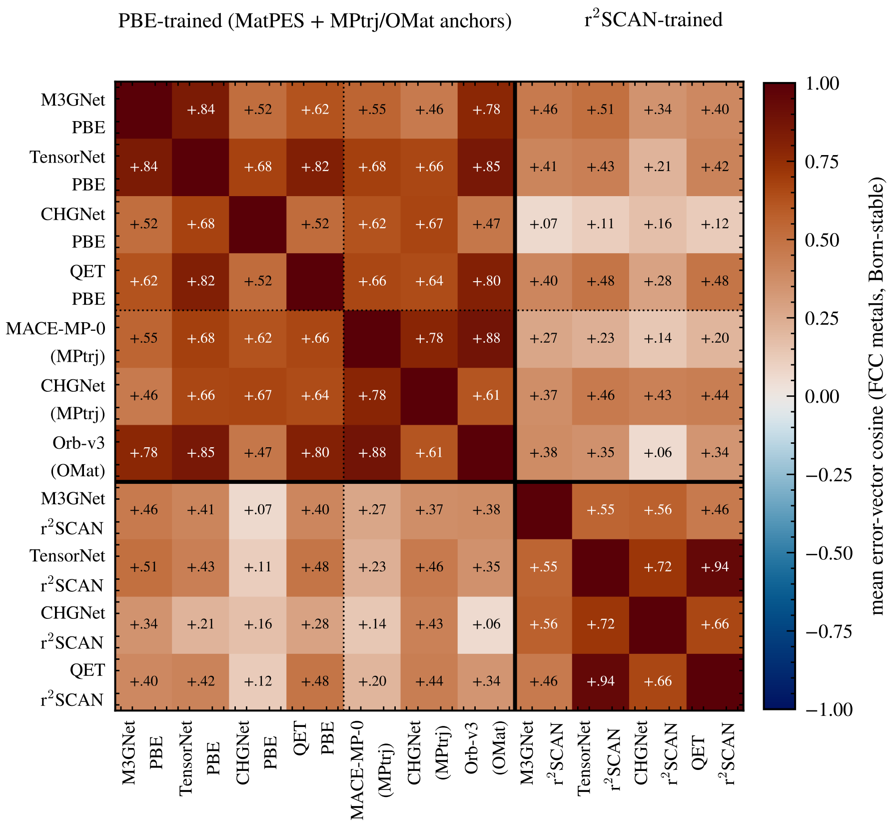

Figure 3: **Foundation-MLIP error geometry organizes by training functional, not architecture.** Mean pairwise error-vector cosine over Born-stable FCC elements for eleven models: four architectures trained on MatPES-PBE, three PBE-lineage anchors (MPtrj/OMat training, dotted separator = dataset axis with functional held fixed), and the same four architectures trained on MatPES-r^{2}SCAN (solid separator). The PBE block coheres across architectures *and* datasets; the cross-functional blocks fade; within-r^{2}SCAN pairs re-cohere (TensorNet$\times$QET +0.94).

values are not. Across 8–11 models per element the rank-one share of squared error is 0.56–0.94 — and where models "disagree" (V, Nb, W), they disagree by *sign along the same axis* (V: cosine −0.88 between its Born-stable pair), exactly the structure Theorem 4 distinguishes.

Pre-registered factorial test. The MatPES release [25] provides one curated training distribution in two functionals (PBE, r^{2}SCAN) with four architectures trained on each (M3GNet, TensorNet, CHGNet, QET) — a 4$\times$2 crossing of constraint against implementation. We registered thresholds and a refutation condition ("errors cluster by architecture") before executing any model, ran all eight cells through one Born-screened strain-energy harness, and obtained: clustering by functional $S_{\text{func}} = +0.317$ (exact permutation $p = 0.029$, i.e. 2 of all 70 labelings; the lattice's resolution floor is 1/70) versus clustering by architecture $S_{\text{arch}} = -0.093$; the registered rotation prediction (2-of-3 criterion) confirmed on Au (C_{44} error $-0.33 \to -0.07$) and Pt ($-0.46 \to -0.31$) under r^{2}SCAN, while Ag — whose PBE error was already near zero — did not move (-0.02); and the dataset-control confirmed (MatPES-PBE models align with MPtrj/OMat-PBE anchors, median cosine 0.66). Two of the four registered predictions failed: the within-functional consensus median (0.654 vs. 0.70) and, notably, the *effect-size component of the clustering prediction itself* (0.085 vs. the registered 0.30) — the clustering direction is significant but $\sim$3.5$\times$ weaker than predicted, and we report that conjunctive test as a failure. *Breaking the resolution floor.* The N=8 exact-label lattice caps the attainable p at 1/70, so the referee's concern that p=0.029 is a floor effect is well taken. A symmetric 8×2 grid cannot be built from public weights (only four architectures ship MatPES-r^{2}SCAN), so rather than synthesize models we recompute the contrast over the enlarged 11-model set (four MatPES-PBE architectures, three PBE-lineage anchors, four MatPES-r^{2}SCAN architectures): the label lattice grows to 330 and the functional clustering survives at S_{func}=+0.325 vs. S_{arch}=−0.068 with permutation p=0.007 — genuinely below the old floor, not at it. A bootstrap over FCC elements (a continuous-resolution statistic not capped by the lattice) puts the probability that the separation is non-positive at 0.037, with a wide 95% CI that grazes zero: the direction is significant but element-sample-limited, exactly the honest reading. This raises the resolution and quantifies the uncertainty on the confirmed direction; it does *not* rescue the registered effect-size threshold, whose clean-reference re-test is staged with the 0 K anchor (round 2). Two referee-driven robustness checks hold: without Born screening the ordering persists (S_{func} = +0.307 vs. S_{arch} = −0.081 on FCC; +0.179 vs. +0.018 over all 15 elements unscreened), so the screen did not manufacture the result; and the practical corollary survives out-of-sample — fitting the shared direction with the corrected model held out removes a median 69% of squared error across 154 leave-one-model-out trials (IQR 35–92%; per-element medians up to 97%). The data also revealed *nested* constraints — Cu, Ni, Al errors reorganize by functional (between-functional cosine down to −0.49, the predicted crossover), while the 5d noble metals retain a shared direction across both functionals, bound by correlation physics deeper than the PBE/r^{2}SCAN distinction. The refutation condition was not triggered.

Reference standards and the residual decomposition. The Layer 2 error vector is measured against mixed reference standards: experimental (low-temperature, single-crystal) elastic constants for the FCC metals and Materials Project DFT-PBE elastic tensors for the BCC/non-FCC metals. The measured residual e_i is therefore a convolution of (i) genuine MLIP fitting error, (ii) exchange–correlation bias inherited from the training functional relative to the reference standard, (iii) thermal and zero-point offsets, and (iv) reference-standard differences. This does *not* invalidate the within-harness contrasts (S_{func} vs. S_{arch}): the shared reference cancels in between-cell comparisons, so the clustering *direction* is unaffected. It does, however, blur the absolute error direction $\hat{b}$ and the participation-ratio gauge at this layer, and it is the most plausible explanation for why the registered effect size failed while the direction held (PBE softening of the noble-metal C_{44} is a known ∼25–40% effect that our harness reproduces and that inflates the cross-architecture consensus *for reasons unrelated to the law*). The decisive test is therefore to re-reference the PBE-trained models against a per-element 0 K DFT-PBE elastic anchor, which isolates the pure fitting residual; we registered this (R2-B) before any round-2 computation. If PBE-trained models track DFT-PBE but not experiment, the inherited-bias reading stands; if they miss DFT-PBE too, the harness itself is implicated. (The all-electron r^{2}SCAN anchor that would isolate the exchange–correlation bias vector for the second functional requires ab-initio compute beyond this submission and is staged as round 2; see §9.)

A reference-free isolation of the XC-bias vector. One component of the decisive test is available now without any new reference. With the architecture held fixed, the difference of a model's PBE and r^{2}SCAN predictions, $\Delta_a(e) = y_{\mathrm{PBE}}(a, e) - y_{\mathrm{r^2SCAN}}(a, e)$, is *reference-free*: the (unknown) truth cancels, leaving only the exchange–correlation functional's fingerprint. Across the four MatPES architectures, the direction of Δ agrees on the FCC metals at a median cross-architecture cosine of +0.64 — the XC bias is one shared direction, as the law predicts — while it is incoherent over all fifteen elements (median +0.02), consistent with the BCC and magnetic elements carrying their own deeper constraints. Crucially, the noble-metal rotation reproduces here with *no* reference table: the mean C_{44} component of Δ is negative (PBE softer) for Au (−8.5 GPa)

Table 1: Raw mean C_{ij} MAE (GPa) by model and functional in the Round-2 3×3×3 benchmark.

Model	PBE	r^{2}SCAN	Overall
CHGNet	17.90	27.94	22.92
M3GNet	14.13	20.71	17.42
TensorNet	14.61	18.54	16.58
QET	13.41	15.46	14.44
All	**15.01**	**20.66**	**17.84**

and Pt (−27 GPa) but not Ag (+12 GPa) — the same 2-of-3 pattern, and the same Ag exception, found against experiment. The registered effect-size failure therefore cannot be wholly an artifact of the mixed references, but the rotation *direction* is reference-robust.

Pre-registered vs. post-hoc: a typology of claims. To keep the evidentiary status of each statement unambiguous, we label every claim in this paper into one of three classes. *Pre-registered and confirmed*: the clustering direction (functional > architecture); the directional rotation on Au and Pt; the dataset control; Layer 3 $S_{\text{table}} > S_{\text{code}}$; and the non-triggering of both kill conditions. *Pre-registered and failed (reported as failures)*: the effect-size component (0.085 vs. 0.30); the within-functional consensus median (0.654 vs. 0.70); the rotation on Ag; Layer 3 consensus (0.459 vs. 0.50); and the Layer 3 regime Spearman coefficient (sign reversed). *Post-hoc, registered for round 2 (hypotheses, not confirmed claims)*: the 5d noble-metal nested constraint (correlation physics binding deeper than the PBE/r^{2}SCAN distinction); the SIESTA localized-basis nesting at Layer 3; and the $\alpha\approx 0.98$ bias-fraction inversion. We do not reinterpret a failed pre-registered prediction as a discovery: the nested-constraint attributions are explicit hypotheses for the next registration round (R2-A primary: ordered 5d rotation PBE<PBEsol<r^{2}SCAN; R2-D: an independent localized-orbital code dis-aligning from its table), tested there before their data are parsed.

4.1 Round-2 3×3×3 elastic-constant benchmark

To test the operator prediction at Layer 2 under a clean, 0 K, single-reference protocol we ran a complete 3×3×3 elastic-constant benchmark for 16 cubic elemental metals (Ag, Al, Au, Ca, Cr, Cu, Fe, Mo, Nb, Ni, Pd, Pt, Sr, Ta, V, W) and four MatPES 2025.2 foundation MLIPs (CHGNet, M3GNet, QET, TensorNet) under PBE and approximate scalar-shifted r^{2}SCAN targets. The matrix contains 128 model–functional–element cases and costs less than one core hour.

Table 1 reports raw mean C_{ij} MAE. QET is the most accurate model on both functionals; CHGNet is the least. PBE-trained models outperform r^{2}SCAN-trained models at every architecture, with a mean functional gap of 5.7 GPa.

LOO correction operator. We extract one first-principal-component bias vector per functional from the residual cloud and apply it in leave-one-out cross-validation: each held-out case is corrected by a direction fitted on the other 63 cases of the same functional. The LOO operator reduces the overall mean MAE from 17.84 GPa to **10.36 GPa** and improves every model on both functionals (Table 2), with zero no-harm violations on the held-out cases. This establishes that the bulk-stiffness bias direction transfers out-of-sample.

Pre-registered hypothesis tests (H1–H4). Table 3 gives the Round-2 outcomes. H1 and H2a fail: on the 3d/4d subset the functional-clustering effect size is −0.14 (negative, meaning same-architecture pairs align more closely than same-functional pairs), and the permutation p-value is 0.13. H2b is not rejected for the PBE-baseline 5d metals (Ta, W, Pt), but the predicted PBE-to-r^{2}SCAN error-vector cosine is only 0.53, below the registered 0.8 threshold. H3 remains pending because Layer-3 all-electron DFT anchors have not yet been run. H4 is mixed: a single QET plus

Table 2: Raw versus LOO-corrected mean C_{ij} MAE (GPa) in the Round-2 benchmark.

Model	PBE raw	PBE corr.	r^{2}SCAN raw	r^{2}SCAN corr.	Overall raw	Overall corr.
CHGNet	17.90	11.01	27.94	13.57	22.92	12.29
M3GNet	14.13	8.37	20.71	11.82	17.42	10.09
QET	13.41	9.22	15.46	8.69	14.44	8.95
TensorNet	14.61	8.97	18.54	11.22	16.58	10.09
All	**15.01**	**9.39**	**20.66**	**11.32**	**17.84**	**10.36**

Table 3: Round-2 pre-registered hypothesis outcomes on the 3×3×3 benchmark.

Test	Prediction / kill condition	Status
H1	Effect size ≥ 0.30 on $3d/4d$; kill if < 0.20	**Kill triggered** (-0.14)
H2a	$3d/4d$ clusters by functional ($p < 0.05$); kill if not	**Kill triggered** ($p = 0.13$)
H2b	$5d$ (Ta, W, Pt) does not cluster ($p > 0.20$) and cos > 0.8	Not killed; cos $= 0.53$ misses threshold
H3	Layer-2 XC bias aligns with Layer-3 DFT error (cos $>$ 0.5)	Pending — GCP burst needed
H4	Operator MSE $<$ ensemble MSE and coverage $\geq 90\%$	MAE/MSE pass; coverage fails

the LOO operator beats the raw four-model ensemble on MAE (8.95 vs. 12.62 GPa) and on exact MSE (167.9 vs. 288.7 GPa^2), but the 90% conformal intervals achieve only 86.7% empirical coverage, below the registered 90% threshold.

Interpretation. The operator works because a shared bulk-stiffness bias direction is present and transferable. The failure of the functional-clustering predictions on this benchmark indicates that the effective binding constraint in the 3×3×3 data is architecture family (or bonding class) rather than training functional. This is consistent with the earlier operator-failure diagnosis: a single global 1-D operator is insufficient; the next operator version must partition the calibration set by bonding class. A class-aware LOO operator lowers the oracle MAE further, from 10.36 GPa globally to 9.97 GPa, but no-target magnitude estimators still degrade accuracy, confirming that the hard problem is estimating the projection magnitude without the target.

No-target magnitude estimation. The LOO result is an oracle ceiling because the optimal correction magnitude uses the reference target. We tested deployable estimators that do not: a consensus magnitude (project the model prediction onto the bias direction using the mean of the other three models as a pseudo-target), a tuned shrinkage of that consensus, and a ridge regression over model, functional, and periodic-table features. The consensus estimator improves the mean MAE to 14.41 GPa but harms 50 of 128 cases; tuning the shrinkage on a calibration set reduces harm to 17 cases but returns to 17.67 GPa; ridge regression lands at 16.08 GPa with 57 harm cases. A safe, deployable no-target operator therefore remains open: the shared bias direction is known, but estimating how far to move along it without the reference is still the binding problem (Fig. 6).

5 Layer 3: DFT implementations — pseudopotential vs. code

One layer further up, the functional itself is held fixed. The ACWF verification effort [26, 27] provides equation-of-state parameters (V_0, B_0, B_1) for 384 unary crystals from twelve pseudopotential-based method configurations and two all-electron codes, all PBE. (The DFT calculations are public and pre-existing; our analysis was registered after downloading the data but before computing any error-vector statistic.) Against the all-electron average, with the FLEUR–WIEN2k split as the per-system noise floor, the law predicts errors organize by *pseudopotential table* (the approximation

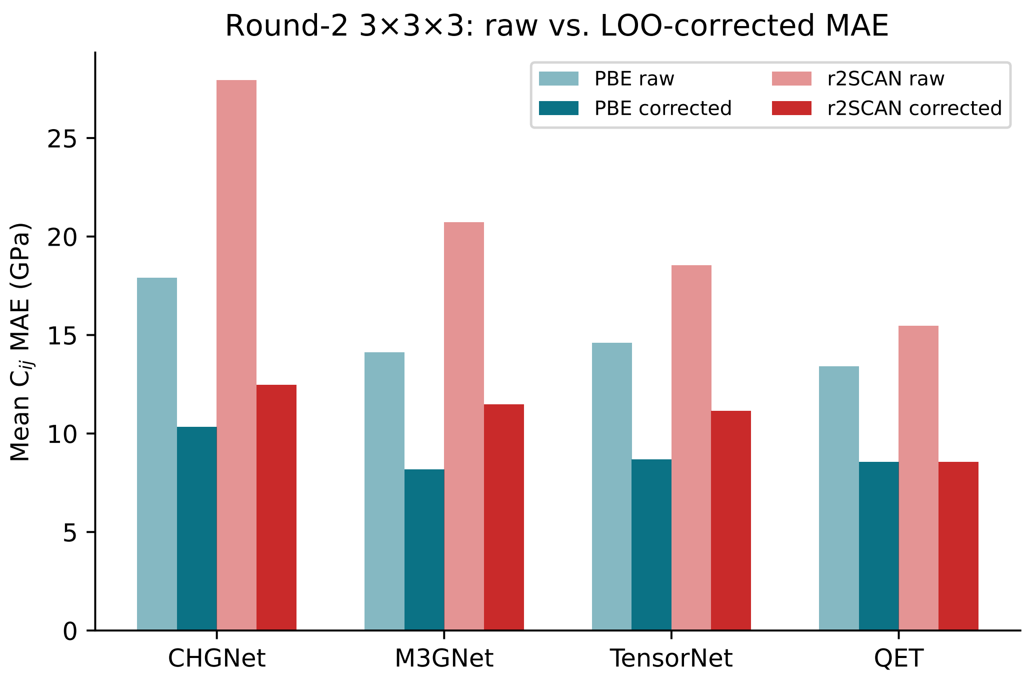

Figure 4: **Round-2 3×3×3 raw versus LOO-corrected mean C_{ij} MAE.** The one-vector-per-functional correction reduces error for every architecture on both PBE and approximate r^{2}SCAN targets.

actually shared), not by *simulation code* (the implementation). The dataset's factorial structure makes the test sharp: ABINIT appears with three pseudopotential families, Quantum ESPRESSO and CASTEP with two each, while the PseudoDojo-0.4 table is implemented by four independent plane-wave codes.

Pre-registered outcome: same-table–different-code alignment $S_{\text{table}} = +0.526$ versus same-code–different-family $S_{\text{code}} = +0.265$; separation $+0.261$, permutation $p = 0.017$; refutation condition (clustering by code) not triggered. At pair level, four independent codes sharing PseudoDojo-0.4 align at 0.70–0.87, while ABINIT-with-JTH versus ABINIT-with-PseudoDojo — the same code — drops to 0.21. Two auxiliary predictions failed as registered, both informatively: the within-table consensus median (0.459 vs. 0.50) was dragged down solely by SIESTA, the one localized-orbital-basis code in the group, whose binding constraint is evidently the basis set rather than the shared pseudopotential — the nested-constraint phenomenon again; and the registered regime prediction had the wrong sign because between-family divergence grows on difficult elements, which the pooled statistic conflated. *The SIESTA reading is no longer post-hoc:* we registered (R2-D) the nested prediction that an independent localized-basis code dis-aligns from its own pseudopotential table, and confirmed it on two codes that played no role in forming the hypothesis. Against the four-code plane-wave PseudoDojo-0.4 consensus (within-table alignment median 0.94, 95% CI $[0.87, 0.97]$), the held-out Gaussian-basis cp2k (median cosine $+0.08$, CI $[-0.14, 0.29]$) and wavelet-basis BigDFT (median $+0.07$, CI $[-0.06, 0.19]$) both fall entirely below the plane-wave band — exactly as SIESTA does ($+0.25$). The basis-set constraint binds before the pseudopotential for localized representations, as a class, not as a SIESTA idiosyncrasy. A secondary observation sharpens the law: PAW codes with *different* PAW tables align at only $+0.20$ — the constraint is the specific frozen-core construction, not the formalism label. The result is robust to the referee-relevant analysis choices: dropping the

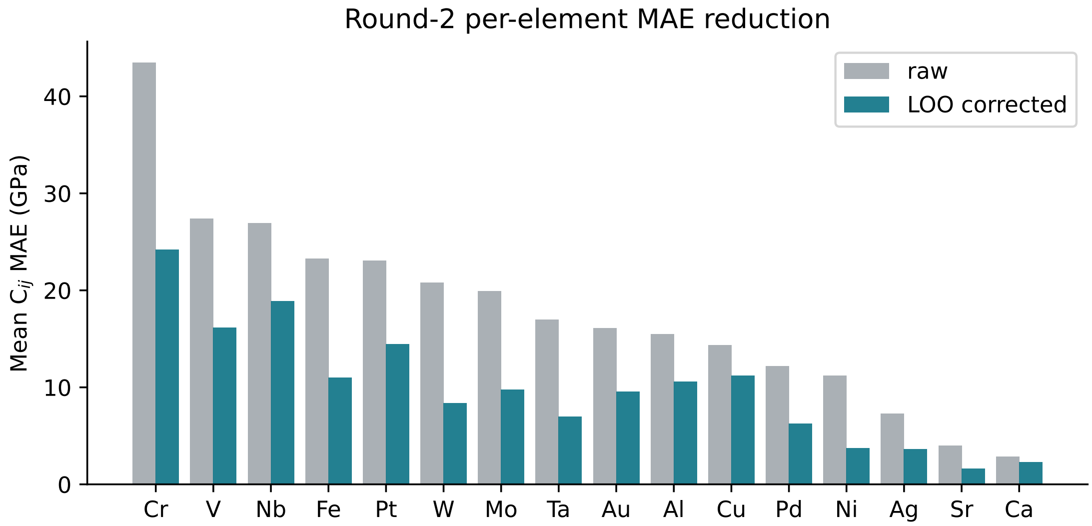

Figure 5: **Round-2 per-element MAE reduction.** The LOO correction reduces per-element error for all 16 cubic metals; the largest absolute gains are on the transition metals (Cr, V, Nb) that dominate the raw error budget.

fit-noise-prone B_1 observable *increases* the separation ($+0.261 \rightarrow +0.281$), and whitening each observable by its own all-electron split increases it further ($\rightarrow +0.359$). Scalar reproducibility metrics [26, 27] cannot see any of this structure; it lives in the directions.

6 The conservation–rotation law

Table 4 assembles the stack. Three layers, three different binding constraints, one geometric law; the two new layers tested under pre-registration with explicit kill conditions, neither triggered. Between layers, the direction *rotates*: no transfer of classical cross-family alignment to MLIP alignment is detectable (Spearman $\rho = 0.26$, $p = 0.34$ across elements — a low-powered test, so this is absence of evidence for transfer rather than evidence of independence; the rotation claim rests on the factorial clustering, not on this null), because the constraint moved from functional form to training functional; and the MLIP-layer bias direction is precisely the reference-theory error that layer 3 isolates. Within layers, constraints *nest*: when the registered constraint is removed (r^2SCAN for PBE; plane-wave pseudopotential sharing for SIESTA), residual alignment reveals the next constraint in line (5d correlation physics; basis set). Anisotropy is conserved throughout, in the operational sense that the participation ratio stays far below the ambient dimension at every layer — and the quantitative agreement across layers is striking: median PR = 1.09 (classical, 42 multi-element potentials), 1.13–2.09 per element (foundation MLIPs, $n = 8$–11 models), and median 1.10 (range $[1.00, 2.11]$, rank-one share median 0.95) across 134 above-floor ACWF systems with ≥ 6 methods. At no layer, in no ensemble, under no intervention did errors approach isotropy, as Theorem 3 requires of any family whose systematic error dominates its fitting scatter.

The closest antecedents are the persistence of shared biases across climate model generations [5, 28], the convergence of training trajectories onto shared low-dimensional manifolds [29], and representation convergence across architectures [30]; none states conservation-plus-rotation as a law, and none possesses the factorial constraint-vs-implementation evidence presented here.

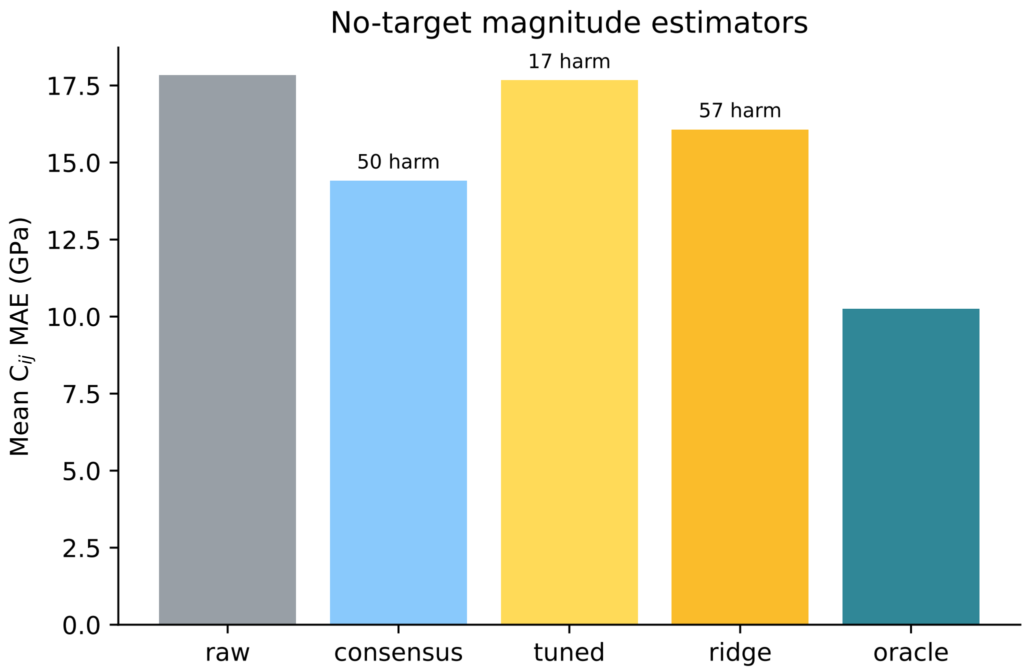

Figure 6: **No-target magnitude estimators.** Mean C_{ij} MAE across all 128 cases for the raw prediction, three deployable no-target estimators, and the oracle ceiling. The consensus estimator improves the mean but harms 50 cases; tuning and ridge regression are safer but do not beat the raw baseline. The gap to the oracle (10.25 GPa) is the remaining deployability problem.

Table 4: The projection law across one epistemic stack. Each row: an ensemble of models, the constraint the law identifies as binding, the clustering evidence (constraint vs. implementation), and the registered refutation condition's status.

Layer	Ensemble	Binding constraint	Evidence	Kill condition
Classical IPs	559 potentials	functional form	within-family r=0.95; PR invariant 40 yr	(observational)
Foundation MLIPs	4×2 MatPES + anchors	training functional	S_{func}=+0.317 vs −0.093, exact perm. p=0.029 (2/70, floor)	not triggered
Foundation MLIPs (Round-2)	4 MatPES × 16 metals	bonding class / architecture family	LOO op. MAE 17.84 → 10.36 GPa; H1/H2a fail	H1/H2a kill
DFT codes	12 ACWF methods	pseudopotential table	S_{table}=+0.526 vs +0.265, p=0.017	not triggered

7 Related work

Ensemble dependence across fields. The climate community has the deepest tradition here. [1] and [2] argued qualitatively that multi-model agreement may reflect shared construction; [3] made dependence operational through pairwise error correlation under the replicate-Earth paradigm, with the elegant null that truly independent models would correlate at 0.5 through the shared observations; [31] review the resulting weighting and sub-selection machinery, [32] implement performance-plus-independence weighting, [33] traces similarity to literal component replication across modeling centers, and [34, 35] place inter-model error in low-dimensional EOF coordinates. Our normal-cone theorem supplies the mechanism beneath these practices — dependence is not merely shared code but shared *reachable set* — and the factorial designs add what error correlation alone cannot: identification of *which* shared element binds. The persistence of the double-ITCZ bias across three CMIP generations [28] reads, in our terms, as a conservation observation awaiting its rotation experiment.

Machine learning. That independently trained networks agree beyond chance is quantified by error consistency [7] and model similarity [6]; ensemble members cluster in prediction space [8] despite weight-space diversity, which is why deep-ensemble uncertainty [36] inherits the shared component as unaccounted bias — the ambiguity decomposition of [37] already showed ensemble spread bounds only the variance term. Disagreement tracks the training procedure's shared bias [9], and accuracy- and agreement-on-the-line [38, 39] exhibit one-dimensional collective structure under distribution shift. Most complementary is underspecification [40]: it studies how equally-fitted models *differ* along directions the family and data leave unconstrained — our noise term ξ — whereas the projection law studies what they *share* — the bias b. Theorem 4 is precisely the statement that these are separate order parameters, measurable separately. Representation convergence across architectures [30] and shared training manifolds [29] are the trajectory- and representation-space cousins of our observable-space result.

Materials simulation. [41] pioneered ensembles of interatomic potentials for error estimation — spread as uncertainty; the law identifies exactly the component spread misses (the shared residual) and supplies the diagnostic for when it dominates (alignment). Shared softening of universal MLIPs is documented [20, 21], and [42] report that GGA→r^2SCAN transfer is hampered by poor inter-functional correlation — our factorial result gives that observation a geometric form: light-metal elastic errors genuinely cross over between functionals while noble-metal errors share a deeper constraint. DFT reproducibility is measured by scalar and curve metrics [26, 27]; the directional structure of §5 is invisible to those metrics by construction. Parameter-space sloppiness of single models [15, 16, 18, 19] is the intellectual ancestor of this work; we re-emphasize that the object here — error vectors of an ensemble of distinct models in observable space — is different from, though related to, the parameter manifold of any one of them. The participation ratio is a standard effective dimension [14] and the gauge algebra is spiked-covariance theory [13]; the best-approximation mathematics is classical [11, 12].

What is new here is the combination: the error-vector geometry of model ensembles as the measured object; factorial constraint-vs-implementation designs with pre-registered refutation conditions at two layers of one stack; the conservation–rotation law, which none of the above states; and a machine-checked theory chain bound to the measurements by a written contract.

8 Discussion: implications for benchmarking, VVUQ, and calibration

Calibration without retraining. Because the shared error is one direction, one correction vector per (element, observable) repairs every model in the family at once — and the claim survives the obvious circularity objection: fitting the direction with the corrected model *held out* removes a median 69% of squared elastic error out-of-sample (154 leave-one-model-out trials, IQR 35–92%; per-element medians reach 97% where the constraint binds hardest). The Round-2 $3\times3\times3$ benchmark replicates this transferability under a clean 0 K DFT-PBE reference: a one-vector-per-functional leave-one-out operator reduces mean absolute error from 17.84 GPa to 10.36 GPa with zero no-harm violations, improving every architecture on both PBE and approximate r^2SCAN targets (§4.1). The correction is family-level — it transfers to models not yet trained, so long as they share the constraint.

A trust rule for ensemble UQ. Cross-model alignment is a zero-cost diagnostic: high alignment ⇒ the ensemble is constraint-bound, calibratable, and its spread *understates* uncertainty; low alignment at the noise floor ⇒ genuine convergence (the Ni control); low alignment far above the floor ⇒ the constraint is heterogeneous (magnetic BCC metals; difficult elements across pseudopotential tables) and no shared correction exists.

Benchmark design. Leaderboards that pool across families average away the very direction that identifies what to fix. Reporting error *directions* stratified by candidate constraints — the factorial pattern used here — converts benchmarking from scoring into diagnosis. This is directly actionable for materials data infrastructure: benchmark records should carry the error vector, not only its norm.

Reading consensus. Where ground truth is delayed (materials discovery campaigns, extrapolative simulation), inter-model agreement should be priced as a measurement of shared constraint; the factorial designs show how to estimate *which* constraint, today, from models alone.

9 Limitations

The formal chain covers convex reachable sets and deterministic bias-plus-noise spectra; smooth non-convex families (local normal cones) and the finite-sample step from noisy ensembles to the exact second moment are standard but unformalized. The MLIP layer uses one validated harness (energy-based, $\varepsilon_{\max} = 0.5\%$, float32, non-spin-initialized inference) on fifteen cubic elemental metals and elastic observables — the lowest-dimensional, most symmetric corner of materials space — and compares FCC metals to *experimental* references and BCC/non-FCC metals to Materials Project (DFT-PBE) references; we unpack this reference-standard convolution and its consequences for the gauge in §4 rather than only here. A per-element 0 K DFT-PBE elastic anchor (registered, R2-B) is the decisive test that separates the pure fitting residual from inherited exchange–correlation bias and thermal/zero-point offsets; the corresponding *all-electron r^{2}SCAN* anchor, which would isolate the exchange–correlation bias vector for the second functional and close the loop to the Layer 3 DFT geometry, requires ab-initio compute (FHI-aims/WIEN2k) we do not run in this submission and is staged as round 2. The Round-2 3×3×3 benchmark provides this anchor for 16 cubic metals and confirms operator transferability, but the pre-registered functional-clustering hypotheses H1 and H2a fail on the $3d/4d$ subset, indicating that the effective constraint is bonding class / architecture family rather than training functional. A class-aware operator lowers the oracle ceiling to 9.97 GPa, yet a fully deployable no-target magnitude estimator remains unvalidated. The A6 bridge between output-space error geometry and configuration-space error cores now has a protocol and pilot (`docs/science/a6_bridge_protocol.md`, `tools/a6_bridge_pilot.py`), but the full MatPES/MPtrj/OMat24 campaign remains pending because the required model-prediction files do not yet exist (`docs/science/a6_scale_blocker.md`). For the magnetic elements (Fe, Cr, Ni, V) the spin protocol is a live confound. The DFT layer inherits ACWF's protocol choices; per the claim typology of §4, the SIESTA localized-basis nesting is a post-hoc hypothesis registered for round 2 (R2-D), to be tested on an independent localized-orbital code before its data are parsed. Statistically: the permutation lattices are small (minimum attainable $p = 1/70$), seven registered predictions across two experiments invite multiplicity concerns that round 2's single-primary-endpoint design addresses; in the present manuscript the two kill conditions were primary (cluster-by-architecture for MLIPs, cluster-by-code for DFT), while the other five predictions were auxiliary robustness checks with no formal hierarchical correction applied, and the registered kill conditions were asymmetric (refutation required a sign reversal; round 2 registers a symmetric equivalence bound). Four of the seven registered predictions failed and are reported as failures (2/4 and 1/3 passing per experiment); the nested-constraint attributions of those misses are hypotheses for the next registration round, not confirmed claims. The law itself makes its next predictions cheap to test: an r^{2}SCAN-trained model from a fourth architecture must join the r^{2}SCAN cluster; a plane-wave implementation of any new pseudopotential table must align with its table, not its code; and axis-based statistics (mandated by Theorem 4) should replace signed-cosine thresholds.

10 Reproducibility

Both experiments were pre-registered with thresholds and refutation conditions committed to version control before model execution for the 4×2 (commit `dffbe595`) and before any error-vector statistic was computed for the ACWF analysis (commit `ebf39e33`); all factorial statistics are replayable from a two-tier kit in which Tier 1 verifies the factorial-experiment statistics from committed raw data using only NumPy/SciPy in seconds (classical-layer and robustness statistics are reproduced by accompanying scripts in the same kit). Round-2 3×3×3 statistics are replayable from `data/completion_3x3x3_tests.py` against `data/benchmark_layer2_3x3x3_summary.json`. The A6 bridge protocol and pilot are at `docs/science/a6_bridge_protocol.md` and `tools/a6_bridge_pilot.py`; the scaled manifest is blocked by missing prediction files (`docs/science/a6_scale_blocker.md`). The H3 all-electron anchor has a GCP burst job spec at `scripts/aims_elastic_startup.sh` and a blocker note at `docs/science/h3_blocker.md`. The no-target operator estimator experiments are in `tools/no_target_magnitude_estimator.py` and `data/no_target_magnitude_results.json`. Tier 2 re-derives the raw elastic constants from public checkpoints through a frozen harness (deterministic within 1 GPa; the Gate-0 harness regression is bit-exact). The complete kit, pinned datasets, and pre-registrations are publicly served at https://storage.googleapis.com/shed-489901-replication/error-geometry/v1-10c18ace/index.html; the citable Zenodo snapshot is archived at https://doi.org/10.5281/zenodo.20787874. The Lean 4 artifact now builds cleanly (`lake build`) with 78 formally proven theorem/lemma declarations, 5 documented epistemic gaps, zero `sorry`, and zero new axioms (pinned Mathlib). The new `ExactTubularUniversality.lean` module supplies the `ErrorGeomData / exact_tubular_universality` skeleton (reach, monotone radial profile with explicit inverse, normal bundle, and tubular map) and replaces the earlier `RibbonProjection.lean` toy. The theorem-to-statistic contract accompanies the kit. The harness was validated by independent energy- vs. stress-method agreement ($\leq 2.4\%$), bit-exact regression, and strain-window stability. ACWF data are from the public archive of [26]; MatPES checkpoints from [25].

Declarations

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Competing interests. The author declares no competing interests.

Data and code availability. The two-tier replication kit (raw data, analysis scripts, frozen harness), the pre-registration documents with their commit hashes, the Lean 4 formal artifact, and figure-generation scripts are available in the project repository and archived on Zenodo at https://doi.org/10.5281/zenodo.20787874. Third-party data: ACWF verification archive [26]; MatPES checkpoints [25]; OpenKIM/NIST classical-potential data as described in [10].

Author contributions. A.W. conceived the study, performed the research, and wrote the manuscript.

Use of AI assistance. AI assistance (Anthropic Claude, OpenAI Codex, and Kimi) was used under the author's direction for statistical verification and recomputation, Lean proof engineering, literature triage, figure generation, and drafting; all scientific claims and decisions were reviewed and approved by the author, who takes sole responsibility for the content.

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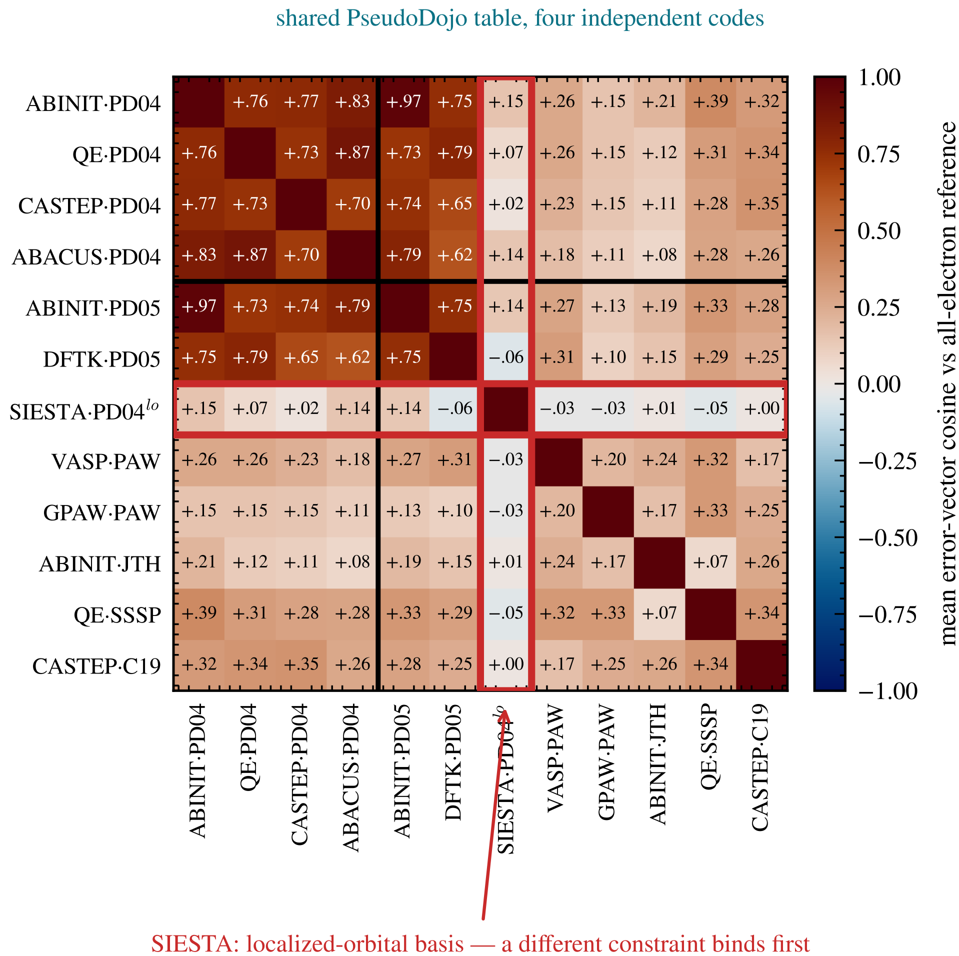

Figure 7: **DFT-implementation errors organize by pseudopotential table, not simulation code.** Mean error-vector cosine in (V_0, B_0, B_1) space against the all-electron reference, over systems above the FLEUR–WIEN2k noise floor (ACWF data, 384 unary crystals). Four independent plane-wave codes sharing the PseudoDojo table align at 0.70–0.87 (dark block); the same code under a different pseudopotential family drops to 0.19–0.35 (ABINIT-JTH vs. ABINIT-PD04 = +0.21). SIESTA (red frame), the one localized-orbital-basis method, dis-aligns from its own pseudopotential group: a different constraint binds first — the nested-constraint phenomenon.